

SealingLid: FDM 3D Printing Technique that Bends Thin Walls to Work as a Lid

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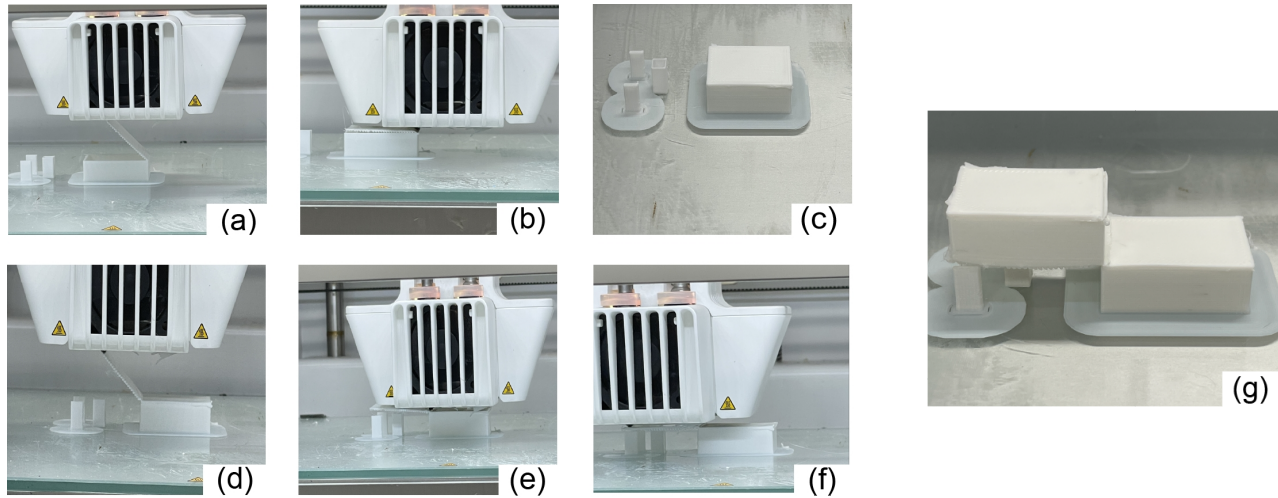


Figure 1: Printing sequence from (a) to (g) with SealingLid. (a) the 3D printer creates a thin slanted wall from an existing edge. A small support structure is also built in advance. (b) the printer head pushes the wall inward and remelts the crease to create an orthogonal corner. (c) additional layers are printed on the lid. (d) the second slanted wall is created outward. (e) the head pushes the wall outward and remelts the crease. (f) layers are printed on the lid. The support structure keeps the lid flat. (g) after the third lid is created, the object, which consists of two hollow rectangles joined together, is finally completed.

ABSTRACT

3D printers inspire user creativity by allowing them to design and create actual 3D objects. Unfortunately, printers often require long printing time and waste a lot of materials. Various techniques have been proposed to alleviate the issues, such as modification of the model and the user's intervention during the printing process. We propose another approach, SealingLid, that creates thin walls and bends them by the printer's head to form lids. It does not require model modification or user intervention and works with simple unmodified FDM 3D printers. A test confirms that the technique reduces the materials used for infill and support structures. Some primitive objects are fabricated to explore the possibilities of the technique.

CCS CONCEPTS

• Human-centered computing → Systems and tools for interaction design.

KEYWORDS

3D printer, FDM, print time, material consumption

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1 INTRODUCTION

3D printers have become popular due to their ease of printing, high customizability with a wide range of materials, and low introduction costs. They inspire the user creativity by offering opportunities to create something they need and want. The FDM technique is the most popular 3D printing technique; it builds layers from the bottom up. However, FDM 3D printers have challenges, such as long printing times and wasting additional material if support structures are needed. The limited speed of nozzle movement is one reason,

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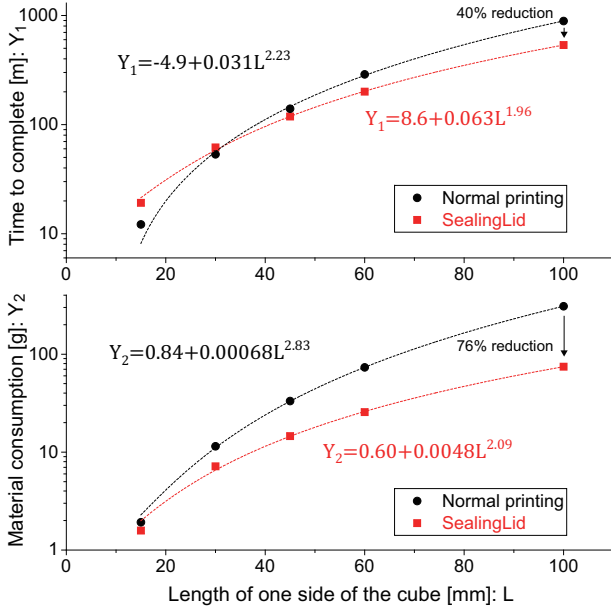


Figure 2: Printing time (upper) and material consumed (lower) when printing cubes. The technique reduces both of them in many cases. The reduction rate depends on the size of the cube. Both indices are approximately proportional to the square of cube length with the technique.

but the other primary reason is the need to build infill and support structures to maintain strength and enable overhangs to be printed.

Techniques have already been proposed to reduce the printing time and material used with FDM 3D printers. Modification or simplification of the model is one of the approaches. For example, WirePrint [6] prints wire models, while Platener [1] modifies the model to replace some surfaces with plates made by a laser cutter. However, they have several drawbacks, such as a loss of detail and the difficulty of testing physical interactions. Generating unique G-code can reduce the time and materials. For example, splitting the model into parts that are printed separately [10], efficient support structures [3], and inclined layer printing to reduce the necessity of support [5]. However, all have some limitations; their improvements and availability are often restricted. User intervention during the printing process works effectively. For example, Substiports [8] uses LEGO blocks as substitutes for support structures, and Flushner [7] replaces flat walls of the object with flat boards cut by a laser cutter. Unfortunately, these techniques conflict with the advantage of 3D printers being able to fabricate objects automatically. Special 3D printers show further possibilities. RemoMaker [2] uses a rotational cuboidal platform, while Xu et al. uses a programmable building platform with dynamically controlled metal pins [9]. While these work effectively, introducing them to casual users would take a lot of work.

2 SEALINGLID

We propose another approach to reduce the printing time and material used. Inspired by Katakura et al.’s work [4], the 3D printer

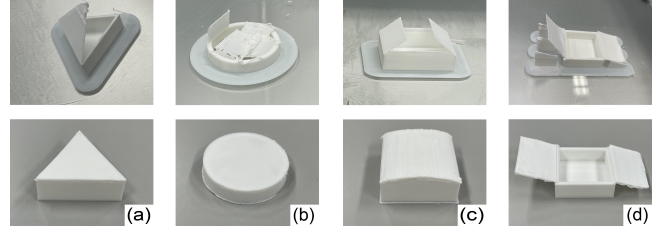


Figure 3: Examples of the technique applications. SealingLid can create (a) triangle, (b) circle, (c) curved inward lids, and (d) outer lids with non-right-angle corners.

nozzle head is used not only for printing but also as a robotic manipulator. The technique works with unmodified FDM 3D printers and does not require model modification or user intervention.

As shown in Figure 1, the technique first prints a thin slanted wall starting from the top on an existing wall without creating a support structure. The printer head physically pushes the wall down to form a lid, and remelts the crease by extruding the filament and seals the lid when possible. The lid works as a foundation on which printing can continue. Pushing the lid inward creates a hollow object, contributing to a reduced infill structure. Bending the lid outward reduces the materials used in creating the support structure. Note that an outward lid needs a small support structure if additional layers are to be printed on the lid. The weight of the layers will cause the lid to sag if no support structure is used. A unique difference from conventional support structures is that the support elements are not fused to the lid, eliminating the need for tedious support removal and cleanup.

3 EXPERIMENT

An experiment was conducted to verify if SealingLid reduces the time and filament used. The experiment printed a cube as the simplest form, with the lid bending inward. Cubes with sides of 15, 30, 45, 60, and 100 mm were fabricated with and without SealingLid to measure the print time and material used. The 3D printer used was an Ultimaker S3, with layer height of 0.15 mm and wall thickness of 1.0 mm. Infill density was set to 20% when printing normally. Ultimaker Cura was used as the slicer, and Ultimaker genuine Touch PLA as the material. The G-code for SealingLid was manually edited by merging several G-codes generated by the slicer.

Figure 2 shows the print times and material consumed. The dotted lines indicate the approximate curves given by $Y = a + b * L^c$, where Y is the time or material used and L is cube dimension. SealingLid reduced the print time and material used in most cases. For example, the time was reduced by 40% and the material usage by 76% for cube dimension of 100 mm. The reduction rate depends on the object’s size; the larger the object, the greater the time and material efficiency. With SealingLid, both indices are approximately proportional to the square of cube length, whereas they are proportional to the powers of 2.23 and 2.83 with normal printing.

4 DISCUSSION

SealingLid works not only with the simplest shape, a cube, but also with other shapes. Figure 3 shows several examples. For example,

triangle and circle lids are also possible. Multiple rectangular lids can be used to form a single circular lid. The technique can create curved lids and generate outer lids with non-right-angle corners. Since these are still primitive shapes, exploring how to apply the technique to complex shapes is necessary.

The current implementation has a few limitations. While the technique achieves the expected reduction rate in material consumption, time-saving is limited. One of the reasons is the slow print speed of the lid. The slicer automatically lowers the print speed to avoid failures when printing the slanted wall. Improving the speed of lids is critical to further reducing overall print times. Moreover, further improvements are needed, such as automating the editing of the G-code, which is currently done manually, and creating a smooth surface for the outer lid, which is often wavy.

5 CONCLUSION

We proposed SealingLid, a 3D printing technique for simple FDM 3D printers that reduces print time and material usage that can be performed solely with an FDM 3D printer. A thin slanted wall is bent inward or outward by the printer head to eliminate infill and support structure. An experiment on printing cubes showed it reduced material usage by up to 76%. Several shapes other than cubes were also produced to confirm the feasibility of the technique.

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